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To cite this article: David Anderson, Matt MacDonald, Paul Ingleby & Nick Brignall (2007) Sightline Jitter Minimization and Shaping Using Nonlinear Friction Compensation, International Journal of Optomechatronics, 1:3, 259-283, DOI: [10.1080/15599610701548837](https://doi.org/10.1080/15599610701548837)

To link to this article: <https://doi.org/10.1080/15599610701548837>



Published online: 10 Sep 2007.



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SIGHTLINE JITTER MINIMIZATION AND SHAPING USING NONLINEAR FRICTION COMPENSATION

David Anderson,¹ Matt MacDonald,² Paul Ingleby,²
and Nick Brignall²

¹Department of Aerospace Engineering, University of Glasgow,
Glasgow, Scotland

²SELEX Sensors & Airborne Systems, Crewe Toll,
Edinburgh, Scotland

Friction-induced sightline jitter significantly degrades the image resolution and detection range of a stabilised optical system. Therefore, any controller that can reduce jitter levels in the stabilisation sub-system will have a significant impact on overall electro-optic system performance. This article presents the results of an investigation into several friction compensation models applied to the validated model of an in-service electro-optic turret. The turret test harness, system identification software and friction measurement techniques used in the validation procedure—experimental transfer function analysis—are presented. A worst-case approach was used in setting the sensor noise and base motion acceleration levels. This test data was then used to validate a mathematical model of the turret elevation axis for use in off-line design and tuning of six friction compensation controllers. Three types of friction compensator model were investigated; a linear Kalman filter, an extended Kalman filter with a static friction model and an extended Kalman filter using a dynamic friction model. Additionally, two controller architectures were used. All six controllers were shown to significantly reduce jitter levels overall, but a new controller architecture was shown to also further reduce image degradation due to smearing.

INTRODUCTION

In high-performance electro-optic (EO) systems, high frequency deviation of the line-of-sight (LoS), known as sightline *jitter*, is dominated by friction in the pointing and stabilisation servomechanism (Netzer 1982; Rue 1969, 1974). Jitter is undesirable as it significantly deteriorates overall EO system performance by both degrading optical transmission in passive systems such as forward-looking infra-red (FLIR) and reducing useful energy on target in active, laser delivery systems, where pointing accuracy is of paramount importance, such as next-generation targeting and directed infra-red countermeasures (DIRCM) systems. It is in the passive imaging channel, however, that the effects of jitter are most important.

To quantify the effect of jitter on image quality it becomes necessary to consider the Modulation Transfer Function (MTF) of the optical train (Johnson 1993). The MTF is the linear spectral representation of optical transmittance as a function of

Address correspondence to David Anderson, Department of Aerospace Engineering, James Watt Building, University of Glasgow, Glasgow G12 8QQ, Scotland. E-mail: d.anderson@aero.gla.ac.uk

NOMENCLATURE

A, B	state space model system and control matrices	$\hat{\mathbf{x}}^-(k)$	a priori state estimate at the k^{th} time step
C, D, G	state space measurement, direct feedthrough and process noise matrices	$\mathbf{w}, \mathbf{v}$	stochastic process and measurement noise
E, H	controller measurement and direct matrices	$\mathbf{X}_{\text{si}}, \mathbf{X}_{\text{pi}}$	stepped-sine and periodic random noise stimulus signals
F_c, F_s	Coulomb friction and Stiction levels	z	Dahl and Lu-Gre model bristle deflection
k	reciprocal of the system inertia	Φ	state transition matrix, controller system matrix
L	Kalman gain	σ_0, σ_1	bristle stiffness and damping term acting on the bristle deflection
P, Q, R	error, process, and measurement noise covariance matrices	σ_2	viscous friction coefficient
r	reference signal	$\omega_i, \omega_b, \omega_s$	inertial, base, and Stribeck velocities
T_f	friction torque	ζ	controller state vector
x, u, y	state, control, and measurement vectors		

spatial frequency. The optical train here includes all lenses and mirrors, as well as the action of the camera system and imaging electronics. Ideally, for optimal detection range and diffraction limited resolution the total scaled MTF should be unity across all spatial frequencies. However, although each component contributes to the overall MTF drop as spatial frequency increases, sightline jitter is the dominant effect (Kopeika et al. 2003). At the operational level, this results in much poorer image resolution—reducing the range at which detection, identification and recognition of the target can be reliably ascertained. Consequently, the reduction of jitter levels is a primary design requirement for all next-generation airborne EO systems.

Jitter degrades the modulation transfer function of the system in two ways – through large-scale frame-to-frame motion and by smearing the image during the sensor integration interval or dwell-time (Hilkert, Bowen, and Wang 1991). Depending on scene content, frame-to-frame motion can often be removed by image re-registration algorithms (Oshima et al. 1989) down to the pixel angular subtense, defined as the solid angle a single pixel presents through the optics. Smearing may only be attenuated electronically by reducing the imager integration time, which will of course reduce the signal-to-noise ratio and consequently the image contrast (Hilkert, Bowen, and Wang 1991). For all gimballed EO systems, which are the vast majority, jitter is dominated by the friction present in the gimbal bearings. Practically, to achieve precision stabilization the optical train must either contain an expensive high-bandwidth fine-mirror-mechanism (FMM) in a Vernier (coarse-fine) arrangement or employ a bespoke friction compensation scheme. For images presented to a human operator, the distinction between frame-to-frame motion and smearing is less relevant as the eye integrates images at around 5 Hz. Automatic target identification systems however mostly rely on feature extraction algorithms that need good contrast in the image to operate effectively and therefore require minimal dwell-time jitter. Current stabilization system design ethos is to consider jitter as a Gaussian process and to design to minimize the overall root-mean-square (rms). It is the contention of the authors that nonlinear

sightline controllers could be designed to both minimize overall jitter levels and simultaneously shape the frequency content of the jitter signal to minimize motion during the imager integration period.

Friction compensation has been investigated using a number of different strategies. First, there is the extension of the classical frequency-domain design techniques using optimal control theory to minimize the compliance transfer function. As we are interested in minimizing the rms of the sightline jitter, one would expect H-infinity synthesis to be appropriate. Indeed, many of the recent sightline control papers have investigated H-infinity methods to shape the closed-loop sensitivity function, (Anderson 2003; Coustal and Michelin 1991). However, as is common in H-Infinity synthesis, it has been shown (Anderson 2003) that a reduction in sensitivity is obtained at the expense of an increase in the noise transfer function (NTF) defined as the closed-loop transfer function from measurement sensor noise to control signal. This is unacceptable because all practical servo control signals are subject to saturation by the motor drive and having excessive noise on the control signal effectively reduces the useful dynamic range. This reduces both the available torque for slew and the linear region of the controller.

A second class of techniques utilises a model-based approach, (Yoichi 1988; Johnson 1974). Here, an accurate model of the plant dynamics (excluding the friction model) is fed the same control signal as the true plant and the outputs are compared. An implicit assumption is that friction torque dominates the model error and that this signal, appropriately scaled, provides an accurate measure of the friction torque. Although successful in some simple cases, this approach is inherently flawed as plant uncertainty and exogenous disturbances will significantly distort the friction estimate and a simple signal comparison provides no bandwidth guarantees on the estimate. This second problem is of particular importance to high performance systems that require high loop gain and/or have to operate using noisy sensors. Operating on the assumption that both plant uncertainty and sensor noise are predominantly high-frequency effects, the torque estimate is usually subjected to some low-pass filtering. However, this introduces additional lag into the estimate, which already suffers from either a single or double integrator delay due to the path from friction torque to measurement signals in the plant. Overall, the resulting compensator can become extremely complex to design well.

More recent attempts at friction compensation have used a variety of mathematical techniques to estimate the friction nonlinearity, (Vivas, Rubio, and Canudas-de-Wit 2002; Alonge, Di Ippolito, and Raimondi 2004; Hensen 2002; Hongliu and Nair 1998; Feemster et al. 1999; Hensen, van de Molengraft, and Steinbuch 2003). The excellent review papers of Armstrong-Helouvry, Wang and Olsson (Armstrong-Helouvry, Dupont, and Canudas de Wit 1994; Wang, Ge, and Lee 2001; Olsson et al. 1998) contain a thorough description of most widely used approaches, while the paper of Ray (Ray, Ramasubramanian, and Townsend 2001), documents an experimental evaluation of several approaches. Ray uses a simple DC motor model to investigate the relative friction compensation performance of a number of the standard friction models and observers, and is therefore of particular relevance to the current investigation. The most interesting result presented in Ray is that the simple Kalman filter observer outperforms all other non-linear dynamic friction models in every test conducted.

Temperature range requirements, system performance specifications and consequent friction level changes during operation necessitate an observer-based approach to the control of friction. One cannot match a friction model under laboratory conditions and freeze the parameters as friction levels and characteristics change considerably during operation (Dahl 1977). All of techniques discussed thus far have at their heart some type of observer structure, the aim of which is to estimate the friction torque. Broadly speaking, these observer approaches can be grouped as either data-driven techniques such as neural networks or model-based techniques based on dynamic friction models (Dahl and Lu-Gre models). The data-driven models have been shown to produce some exciting results (Hongliu and Nair 1998; Wang, Ge, and Lee 2001), but have two serious shortcomings when considered for deployment on an airborne weapons platform—determinism and stability. The neural network based approaches in particular suffer from this problem, which can be roughly stated as the requirement for certification that the designer can demonstrate that safety, reliability, and performance of the controller will be maintained throughout all possible excursions in the flight envelope (Moir and Seabridge 2006). For these reasons in particular, this research focused on achieving a friction compensation using a combination of non-linear estimation and non-linear, dynamic friction models.

This article describes the validation of an accurate stabilisation model of an in-service EO turret and the use of this model in the design and analysis of new non-linear stabilisation controllers for both attenuation and shaping of sightline jitter. The experimental apparatus, test procedure, model architecture and validation results are described and the three observer models used to identify the friction characteristics are developed. A conventional friction compensation scheme for the turret is designed and then a new approach is presented where the signal pathways are altered slightly to induce different spectral content in the jitter signal. Each controller is then evaluated against overall jitter RMS levels and image quality.

SYSTEM IDENTIFICATION AND MODELING

The EO system used in this investigation is a two-axis gimballed turret, which operationally has the sensor package mounted on the inner gimbal. The base of the outer gimbal is attached to the host platform, which can be either an unmanned aerial vehicle (UAV) or conventional manned aircraft. The gimbals isolate the sensor from platform motion in the azimuth and elevation axes, but are additionally actuated by brushless DC motors in response to pointing and stabilisation commands. A picture of the turret in the test harness with a dummy sensor package installed is shown in Figure 1.

Turret Simulation Model

The axis under investigation is the inner elevation gimbal axis, which carries the EO payload. A schematic of the system model and control configuration is shown in Figure 2.

Construction of a single axis servomechanism model of sufficient fidelity for control law design is relatively straightforward (Rue 1969, 1974; Yoichi 1988). A brushless DC motor under current control provided by a PWM drive provides

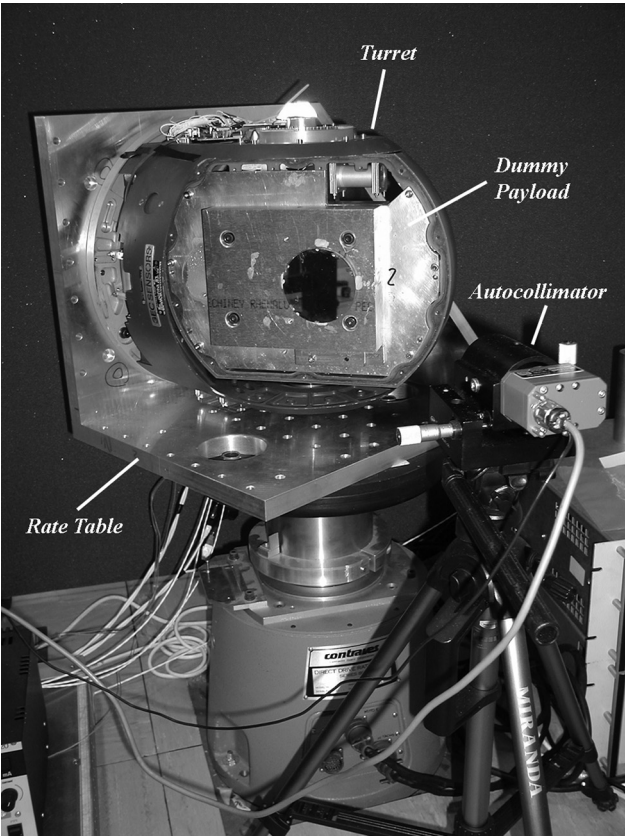


Figure 1. Photograph of the EO turret in the test harness.

actuation of the gimbal axis. The drive is modeled as a transconductance amplifier with an output voltage range of ± 150 V and an integrated 700 Hz current loop under PI control closed around the armature current. Specific commutation dynamics are not modelled as commutation inaccuracies predominantly influence only the high-speed/torque region and stabilisation is a low speed problem. Motor

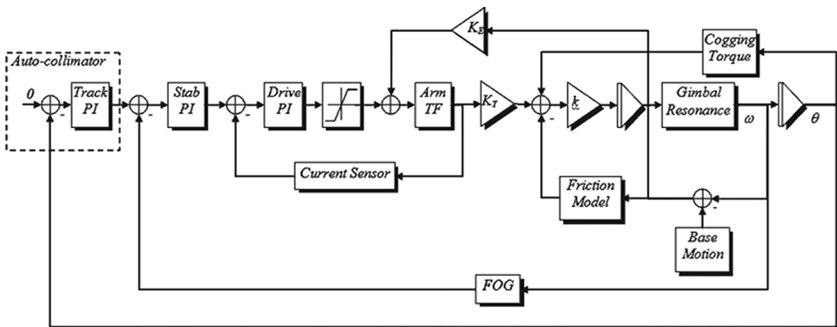


Figure 2. Turret elevation axis block diagram.

back-EMF (K_E) and armature resistance & inductance are included, although their effect is heavily attenuated by the current loop.

Both rigid-body and first torsional resonance are modeled to give the inertial rate and position. On the equipment, base and gimbal inertial rate are measured by fiber-optic gyroscopes (FOG). It is not appropriate to model the optical characteristics of the gyro for a model used in controller design, but the sampling rate (1 kHz) and noise characteristics are included. The output of the FOG drives the stabilisation loop compensator, which is a PI designed for a 25 Hz open-loop bandwidth. Cogging torque is a simple sinusoidal function of armature rotor position at the identified frequency and amplitude. The optical autocollimator provides angular position offset measurement, which drives the track-loop compensator, again PI.

Own-ship motion disturbs the sightline in three ways: through changes to the motor back-EMF induced by the relative velocity across the bearing, friction in the bearings and motion about the roll axis. Motion about the roll axis is controlled via the image-tracking loop, either digitally or using a de-rotation prism and is consequently beyond the scope of this investigation. Details of the various friction models adopted are given below.

Friction model. Friction in the nonlinear plant simulation is estimated using the popular Lu-Gre dynamic friction model (Canudas de Wit et al. 1995). The Lu-Gre model assumes that each contact surface is uneven and that during sliding, the asperities of both surfaces deflect in the same fashion as bristles of a brush. The Lu-Gre model captures finite breakaway stiffness, energy dissipation, hysteretic effects and the Stribeck nonlinearity that governs stick/slip blending and is therefore an excellent model of bearing friction for control design. The Lu-Gre model consists of the following three equations.

$$\begin{aligned} \dot{z} &= (\omega_i - \omega_b) - \frac{|(\omega_i - \omega_b)|}{g(\omega_i - \omega_b)} z \\ T_f &= \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 (\omega_i - \omega_b) \\ \sigma_0 g(\omega_i - \omega_b) &= F_c + (F_s - F_c) e^{-((\omega_i - \omega_b)/\omega_s)^2} \end{aligned} \quad (1)$$

where σ_0 is the bristle stiffness, F_c is the Coulomb friction level, σ_1 is a damping term acting on the bristle deflection, σ_2 is the viscous friction coefficient, F_s is the stiction level, and $\omega_i, \omega_b, \omega_s$ are the inertial, base and Stribeck velocities, respectively.

Test Facility

In the test harness, the sightline is measured using a calibrated commercial-off-the-shelf optical collimator (UDT Model 1020) with a mirror placed in front of a dummy payload mounted on the IG. The optical collimator transmits a narrow laser beam, which is returned by the mirror onto a detector mounted inside the collimator. Off-boresight angle returned by the detector is therefore a direct measure of the sightline deflection of the system under test.

An overview of the stabilisation test is given in Figure 3. The turret is placed on a rate table, which is controlled by a dedicated National Instruments PXI unit.

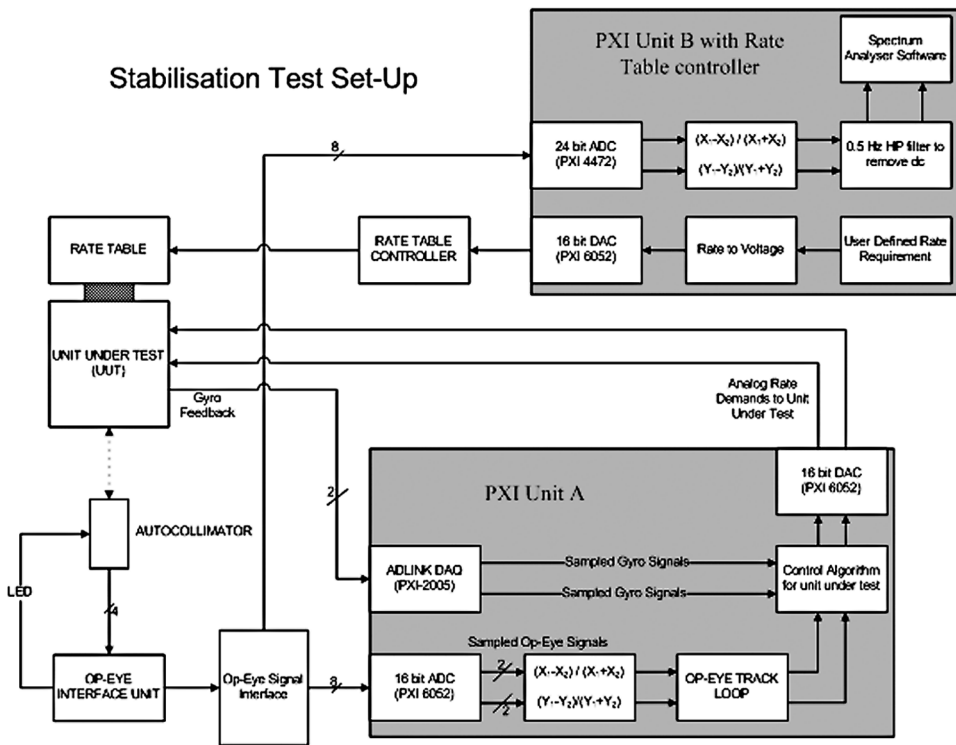


Figure 3. Overview of stabilisation test set-up.

The rate table is used to inject base-motion disturbance into the stabilisation loop of the axis under test. During the servo identification, the outer position loop is not employed. This loop is only introduced to the stabilisation tests. A mirror is mounted on the axis under test and the turret position is monitored using an electronic autocollimator. The position loop ensures the autocollimator is aligned with the mirror in order to obtain a stabilisation measurement. However, the bandwidth of the position loop is very low (< 0.5 Hz) so that it does not interfere with the stabilisation results achieved by the turret.

System Identification

Frequency response. Figure 4 shows the basic set-up for the frequency response tests carried out on the prototype rig. By controlling the stimulus signal and monitoring the signals X_{si} , X_{pi} , and Y_0 we can obtain the transfer function for the closed loop, open loop, controller and plant in a single test. Tests using both a Periodic Random Noise Stimulus and a Stepped Sine Stimulus were completed on both Azimuth and Elevation axis. Figure 5 shows the open loop transfer function for the Elevation axis. The plant demonstrates a characteristic -20 dB/decade gain consistent with a single integrator plant. A major resonance in the plant can then be seen

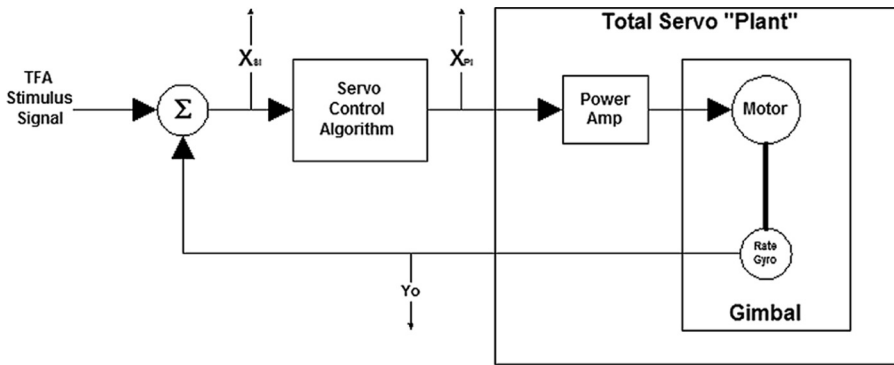


Figure 4. Basic TFA set-up.

at approximately 170 Hz. The large drop in the gain of the plant is accompanied by a large amount of phase loss.

Friction identification. The friction in each axis is measured by driving at a constant low rate and monitoring the output voltage from the controller. From this the current being sent to the motors may be calculated and, using the motors torque constant we have a measure of the friction of the turret. Figure 6 shows the friction measured from the Elevation axis when driven at a constant rate of 0.628 rad/s for a full revolution in both a positive and negative direction.

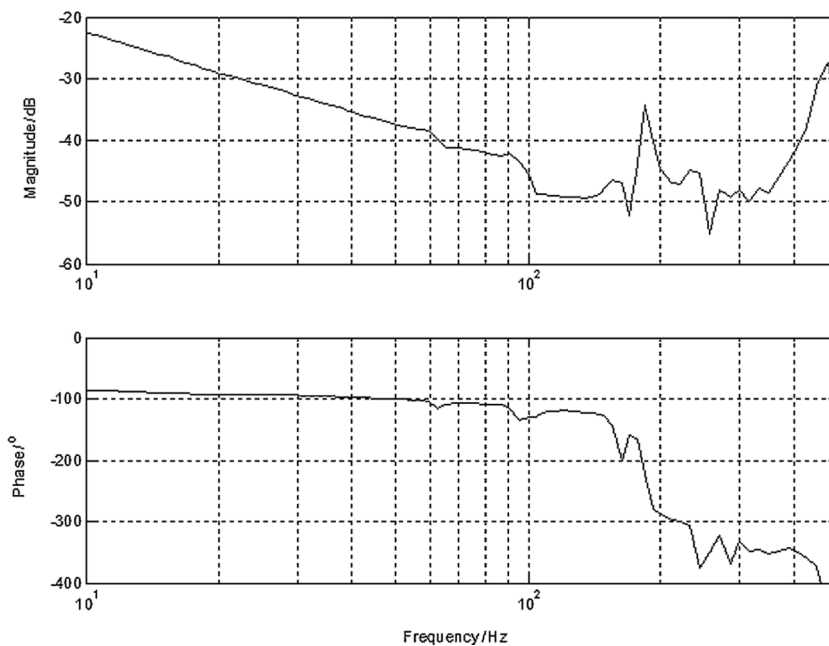


Figure 5. Open loop transfer function for elevation axis.

The friction levels are closer to 0.15 Nm giving a friction to inertia ratio of approximately 1.1. However, note should be taken of the “noise” on the elevation friction plots. This wobble on the friction plot has 79 cycles/rev, which exactly

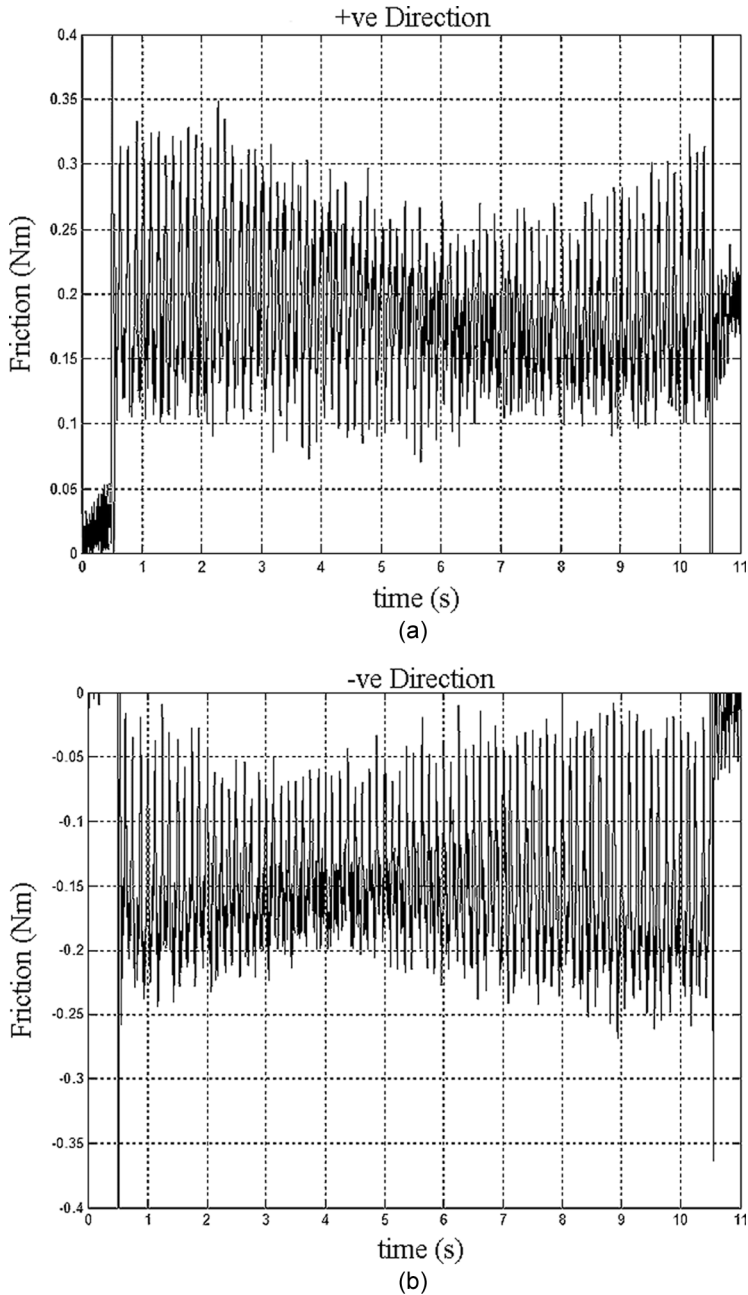


Figure 6. Elevation friction plots for both forward and reverse directions.

matches the torque ripple defined in the elevation motor specification. Further tests driving the Elevation motor with a constant current and monitoring the rotation rate clearly indicate that the fluctuations in friction are due to a cogging torque characteristic exhibited by the motor. The base friction of this axis is therefore a fairly constant value (approximately 0.15 Nm). However, cogging within the motor provides a periodic disturbance torque with an amplitude of ~ 0.1 Nm which is superimposed on the baseline friction.

Validation Test

For the turret model to be used for accurate sightline stabilisation controller design, it must accurately predict the compliance characteristics of the servo. The compliance is defined here as the transfer function from exogenous disturbance torques to sightline angular deflection. Compliance is measured using a standard transfer function analysis technique. Here a series of sinusoidal base inputs are applied to the turret by the rate table and the stabilised sightline response measured by the autocollimator. As the only disturbance torque acting on the system is the bearing friction, the ratio of autocollimator measurement to base motion gives the compliance frequency response function at the base input frequency.

Figure 7 compares the sightline jitter deflection (in μ rad) of the gimbal set measured by the autocollimator against that obtained by simulation under identical test conditions (identical base motion sine-sweeps). The parameters of the Lu-Gre model were selected by prior knowledge of bearing friction characteristics and the results of the friction identification tests under worst-case base inputs. Inspection of Figure 8 clearly illustrates that the simulation model accurately captures the sightline jitter induced by bearing friction. From this result, it is reasonable to

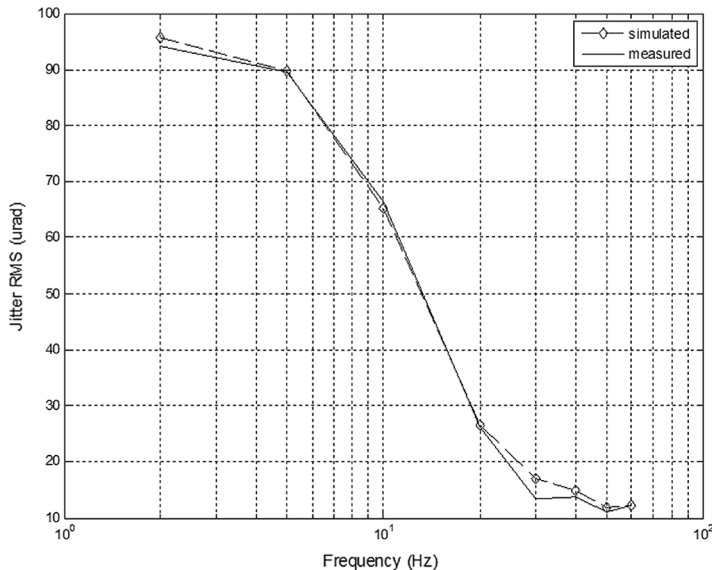


Figure 7. Comparison of simulated and experimental compliance.

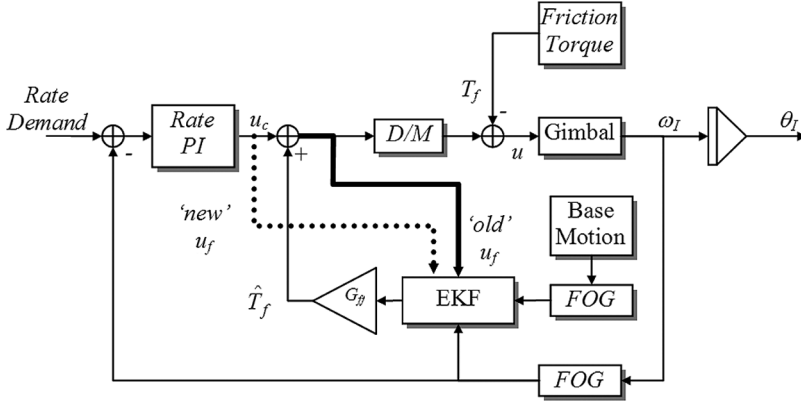


Figure 8. Observer-based control architecture. Note that only one control signal u_f is applied to the EKF at any time.

infer that the EKF controller designs tested on the validated simulation are likely to perform as well on the actual system.

Friction Estimation

Extended Kalman filter. The extended Kalman filter (EKF) is a well-established technique for applying the Kalman filter to non-linear systems. The purpose of the filter is to provide an estimate of the friction torque using nonlinear state propagation models. For the friction models chosen, the nonlinearities can be encapsulated by formulating the model in state-dependant coefficient (SDC) form (Belikov, Kamali, and Friedland 1997; Cloutier 1997; Cloutier and Mracek 1997; Friedland 2001). The system model in SDC form is then,

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}(\mathbf{x})\mathbf{x} + \mathbf{B}(\mathbf{x})\mathbf{u} + \mathbf{G}(\mathbf{x})\mathbf{w} \\ \mathbf{y} &= \mathbf{C}(\mathbf{x})\mathbf{x} + \mathbf{D}(\mathbf{x})\mathbf{u} + \mathbf{v}\end{aligned}\quad (2)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the control input, $\mathbf{y}$ is the measurement (inertial rate for all filters), $\mathbf{w}$ is stochastic process noise, $\mathbf{v}$ is measurement noise and,

$$E(\mathbf{w}) = E(\mathbf{v}) = 0, \quad E(\mathbf{w}\mathbf{w}^T) = \mathbf{Q}, \quad E(\mathbf{v}\mathbf{v}^T) = \mathbf{R} \quad (3)$$

where $E(\cdot)$ is the expectation operator. A linearized model of the friction is obtained at each point along the trajectory and this is used to propagate the error covariance matrix $\mathbf{P}$ and consequently the Kalman gain $\mathbf{L}$.

State prediction (from previous time step):

$$\begin{aligned}\hat{\mathbf{x}}^-(k) &= \Phi_d(k-1)\hat{\mathbf{x}}(k-1) + \mathbf{B}_d\mathbf{u}(k-1) \\ \hat{\mathbf{y}}(k) &= \mathbf{C}_d\hat{\mathbf{x}}^-(k)\end{aligned}\quad (4)$$

Kalman gain computation:

$$\mathbf{P}^-(\hat{\mathbf{x}}^-(k); k) = \Phi_d(\hat{\mathbf{x}}^-(k))\mathbf{P}(\hat{\mathbf{x}}^-(k-1); k-1)\Phi_d^T(\hat{\mathbf{x}}^-(k)) + \mathbf{G}(\hat{\mathbf{x}}^-(k))\mathbf{Q}\mathbf{G}^T(\hat{\mathbf{x}}^-(k)) \quad (5)$$

$$\mathbf{L}(\hat{\mathbf{x}}^-(k); k) = \mathbf{P}^-(\hat{\mathbf{x}}^-(k); k)\mathbf{C}_d^T[\mathbf{C}_d\mathbf{P}^-(\hat{\mathbf{x}}^-(k); k)\mathbf{C}_d^T + \mathbf{R}]^{-1} \quad (6)$$

$$\mathbf{P}(\hat{\mathbf{x}}^-(k); k) = (\mathbf{I} - \mathbf{L}(\hat{\mathbf{x}}^-(k); k)\mathbf{C}_d)\mathbf{P}^-(\hat{\mathbf{x}}^-(k); k) \quad (7)$$

State estimate update:

$$\hat{\mathbf{x}}(k) = \hat{\mathbf{x}}^-(k) + \mathbf{L}(\hat{\mathbf{x}}^-(k); k)[\mathbf{y}(k) - \mathbf{C}_d\hat{\mathbf{x}}^-(k)] \quad (8)$$

where the subscript d denotes discrete models and in particular,

$$\Phi(\hat{\mathbf{x}}^-(k)) = \frac{\partial}{\partial \hat{\mathbf{x}}} \mathbf{A}(\hat{\mathbf{x}}^-(k)), \quad \Phi_d(\hat{\mathbf{x}}^-(k)) = \exp(\Phi(\hat{\mathbf{x}}^-(k))T_s) \quad (9)$$

There are several ways of implementing an EKF with varying degrees of numerical robustness and computational tractability. As the purpose of this investigation was to ascertain the performance of three friction models inside an EKF no complex implementation algorithms (Kalman 1960; Lewis 1992) (such as U-D factorization for example) were employed. It should be noted at this point that the EKF does have some theoretical problems with certain classes of nonlinear system, specifically such nonlinearities that do not preserve the Gaussian probability distribution of the filter input. To overcome this deficiency a number of alternative filtering strategies have been developed such as the signal point or unscented Kalman filter (Julier, Uhlmann, and Durrant-Whyte 2000), which follows the same overall strategy as the EKF and sequential monte-carlo type methods such as the particle filter (Arulampalam et al. 2002; Zaveri 2004). As will be shown later, the EKF is a sufficient estimator for the purpose of the current investigation, but these additional techniques are ideal for future research in this area.

Extended Kalman Filter Models

Coulomb friction. The simplest friction model is that of classical Coulomb friction, given by the following equation.

$$T_f = F_c \operatorname{sgn}(\omega_i - \omega_b), \quad \operatorname{sgn}(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases} \quad (10)$$

This model captures the signal reversal around zero velocity but cannot predict the finite stiffness in the bearing, stiction or the Stribeck effect. However, it is possible to change the Coulomb model slightly to include a crude approximation to finite breakaway stiffness by replacing the $\operatorname{sgn}()$ function with a hyperbolic tangent (Friedland 2001). This smoothes out the switching action and can be tuned to give

a friction–velocity profile very similar to the Dahl model. The SDC representation of the filter model then becomes,

$$\begin{bmatrix} \dot{\hat{\omega}}_i \\ \dot{\hat{\omega}}_b \\ \dot{\hat{F}}_c \end{bmatrix} = \begin{bmatrix} 0 & 0 & -k \tanh[\sigma_1(\hat{\omega}_i(t) - \hat{\omega}_b(t))] \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\omega}_i \\ \hat{\omega}_b \\ \hat{F}_c \end{bmatrix} + \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} \mathbf{u} \quad (11)$$

Similarly, the linearised model is given by,

$$\begin{aligned} & \begin{bmatrix} \dot{\hat{\omega}}_i \\ \dot{\hat{\omega}}_b \\ \dot{\hat{F}}_c \end{bmatrix} \\ &= \begin{bmatrix} -\hat{F}_c(t)k\sigma_1 \text{sech}^2[\sigma_1(\hat{\omega}_i(t) - \hat{\omega}_b(t))] & \hat{F}_c(t)k\sigma_1 \text{sech}^2[\sigma_1(\hat{\omega}_i(t) - \hat{\omega}_b(t))] & -k \tanh[\sigma_1(\hat{\omega}_i(t) - \hat{\omega}_b(t))] \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ & \times \begin{bmatrix} \hat{\omega}_i \\ \hat{\omega}_b \\ \hat{F}_c \end{bmatrix} + \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} \mathbf{u} \end{aligned} \quad (12)$$

Linear integrator chain. The second friction model used in this investigation assumes no a-priori knowledge about the structure of the friction model (Ray, Ramasubramanian, and Townsend 2001). Here, the action of the filter estimates the friction state from the measurements. The only structural information the Kalman filter requires is the depth of integrator chain (friction estimate to process noise input) to adequately capture the dynamic response. The integrator chain approach has the same computational overhead benefits as the simple Coulomb friction model. However, with no friction model structure, in particular no explicit velocity reversal information, it is critical that an acceptable balance is met between adequate tracking performance of the friction torque and insensitivity to measurement noise in the tuning phase of the filter design. Unlike the other two friction models, the integrator chain is linear and consequently has no meaningful SDC form. However, the linear model used in the filter is given in Eq. (13).

$$\begin{bmatrix} \dot{\hat{\omega}}_i \\ \dot{\hat{\omega}}_b \\ \dot{\hat{T}}_f \\ \dot{\hat{T}}_f \end{bmatrix} = \begin{bmatrix} 0 & 0 & -k & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\omega}_i \\ \hat{\omega}_b \\ \hat{T}_f \\ \hat{T}_f \end{bmatrix} + \begin{bmatrix} k \\ 0 \\ 0 \\ 0 \end{bmatrix} \mathbf{u} \quad (13)$$

Dahl friction. Neither the simple Coulomb friction nor the integrator chain models explicitly account for any of the second-order observed friction nonlinearities such as finite breakaway stiffness or the Stribeck effect. To capture these effects, a dynamic friction model embedded inside the EKF is required. The Dahl model (Dahl 1977) captures the first of these, the bearing friction stiffness as follows.

$$\dot{z} = \sigma_0(\omega_i - \omega_b) - \frac{\sigma_0|(\omega_i - \omega_b)|}{F_c} z \quad (14)$$

where z is the internal state of the friction model and σ_0 is the model stiffness parameter (note the similarity between the structure of the Dahl model and the Lu-Gre model). Ignoring any viscous friction terms, $T_f = z$ in the Dahl formulation. For most EO applications with the necessary bearing pre-load to avoid excessive misalignment of the optics, a bearing friction stiffness of $\sigma_0 = 10^5$ has been found to yield accurate results. However, the high bearing pre-load values means care must be taken when implementing the Dahl model inside the EKF plant. The state-dependant nature of the dynamics of the Dahl model can quickly induce numerical instability, causing difficulties for the numerical integration or discretisation routines. Rearranging Eq. (14) it is obvious that the Dahl model is simply an ordinary first-order differential equation with time-varying (state dependent) coefficients.

$$\dot{z} + \frac{\sigma_0 |(\omega_i - \omega_b)|}{F_c} z = \sigma_0 (\omega_i - \omega_b) \quad (15)$$

Assuming for the moment that the input $(\omega_i - \omega_b)$ is held constant, the Dahl model has a single pole at

$$z = -\frac{\sigma_0 |(\omega_i - \omega_b)|}{F_c} \quad (16)$$

This pole lies in the range $[0, -\infty)$ depending on the amplitude of $|(\omega_i - \omega_b)|$. However, we are principally interested in the region where $\omega_i \approx \omega_b$ which is around the velocity reversal. The large value of σ_0 leads to significant pole excursions even when $(\omega_i - \omega_b)$ is small and thus dramatically reduces the time constant. To mitigate integration problems and should be “frozen” when $|(\omega_i - \omega_b)| > |(\omega_i - \omega_b)|_{\max}$ as z quickly settles at the Coulomb friction level. The Dahl model in SDC form is then

$$\begin{bmatrix} \dot{\hat{\omega}}_i \\ \dot{\hat{\omega}}_b \\ \dot{\hat{z}} \\ \dot{\hat{F}}_c \end{bmatrix} = \begin{bmatrix} 0 & 0 & -k & 0 \\ 0 & 0 & 0 & 0 \\ \sigma_0 & -\sigma_0 & -\frac{\sigma_0 |\hat{\omega}_i(t) - \hat{\omega}_b(t)|}{\hat{F}_c(t)} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\omega}_i \\ \hat{\omega}_b \\ \hat{z} \\ \hat{F}_c \end{bmatrix} + \begin{bmatrix} k \\ 0 \\ 0 \\ 0 \end{bmatrix} \mathbf{u} \quad (17)$$

where the fourth state is an estimate of the Coulomb friction level. The linearization of this model for the error covariance update is given by

$$\begin{bmatrix} \dot{\hat{\omega}}_i \\ \dot{\hat{\omega}}_b \\ \dot{\hat{z}} \\ \dot{\hat{F}}_c \end{bmatrix} = \begin{bmatrix} 0 & 0 & -k & 0 \\ 0 & 0 & 0 & 0 \\ \sigma_0 \left(\frac{1 - \text{sgn}(\hat{\omega}_i(t) - \hat{\omega}_b(t)) \hat{z}(t)}{\hat{F}_c(t)} \right) & -\sigma_0 \left(\frac{1 - \text{sgn}(\hat{\omega}_i(t) - \hat{\omega}_b(t)) \hat{z}(t)}{\hat{F}_c(t)} \right) & -\frac{\sigma_0 |\hat{\omega}_i(t) - \hat{\omega}_b(t)|}{\hat{F}_c(t)} & \frac{\sigma_0 |\hat{\omega}_i(t) - \hat{\omega}_b(t)| \hat{z}(t)}{\hat{F}_c(t)} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\omega}_i \\ \hat{\omega}_b \\ \hat{z} \\ \hat{F}_c \end{bmatrix} + \begin{bmatrix} k \\ 0 \\ 0 \\ 0 \end{bmatrix} \mathbf{u} \quad (18)$$

Friction Compensation

Most, if not all friction compensation techniques follow a standard disturbance estimation architecture (Canudas de Wit et al. 1995; Feemster et al. 1999; Olsson et al. 1998; Ray, Ramasubramanian, and Townsend 2001). The principle of signal superposition is invoked such that should a suitable friction observer be available, superimposing a negation of the estimated friction torque onto the rate loop control signal will improve stiffness (Armstrong-Helouvry, Dupont, and Canudas de Wit 1994; Ray, Ramasubramanian, and Townsend 2001; Nuninger, Balaud, and Kratz 1997). This is shown in Figure 9 for two different observer architectures.

The friction torque is assumed an exogenous disturbance in this case to avoid introducing unnecessary complication to the analysis. An additional gain term G_{ff} has been placed on the friction torque estimate signal, which for the moment is set to unity but will become significant later. To understand the compensation process, consider the state-space model of the closed-loop transfer function. To simplify the analysis, the linear version of the EKF is assumed but, as will be shown in the simulation results of the next section, this restriction does not compromise the following exposition. Define the state-space models of the plant, controller and Kalman filter as follows.

System:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{Ax} + \mathbf{Bu} \\ \mathbf{y} &= \mathbf{Cx} + \mathbf{D}\omega_b\end{aligned}\tag{19}$$

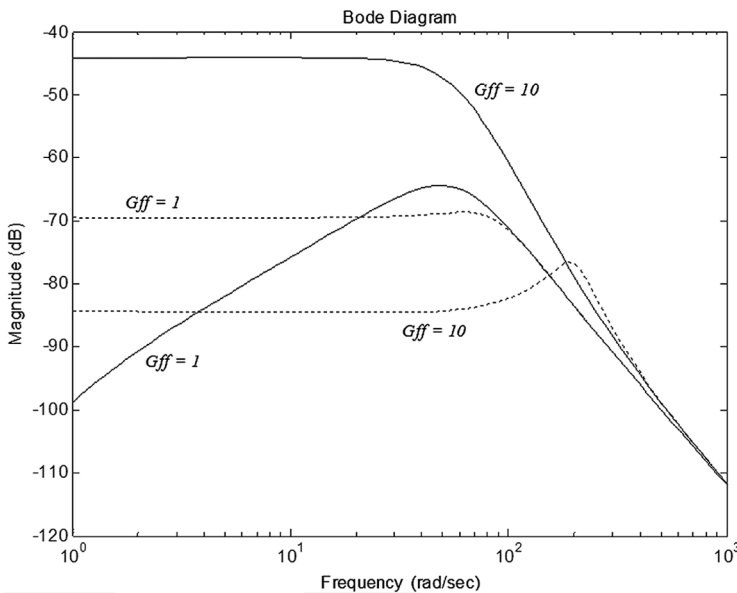


Figure 9. A comparison of the old (solid) and new (dashed) control architectures for two values of feedforward gain G_{ff} . It is clear that the new architecture improves performance for frequencies above 5 rad/s for this example.

Controller:

$$\begin{aligned}\dot{\zeta} &= \Phi\zeta + \Gamma(r - \mathbf{C}^0\mathbf{y}) \\ \mathbf{u}_c &= \mathbf{E}\zeta + \mathbf{H}(r - \mathbf{C}^0\mathbf{y})\end{aligned}\quad (20)$$

EKF:

$$\begin{aligned}\dot{\hat{\mathbf{x}}} &= \mathbf{A}_f\hat{\mathbf{x}} + \mathbf{B}_f\mathbf{u}_f + \mathbf{L}(\mathbf{y} - \hat{\mathbf{y}}) \\ \hat{\mathbf{y}} &= \mathbf{C}_f\hat{\mathbf{x}}\end{aligned}\quad (21)$$

The plant input signal is given by,

$$\mathbf{u}_f = \mathbf{u}_c + G_{ff}\mathbf{C}_f^0\hat{\mathbf{x}} \quad (22)$$

The state, control and measurement matrices have their usual meanings and $\mathbf{C}_0, \mathbf{C}_f^0$ are signal selection matrices used to pick-off the appropriate elements from the state vector. Assuming for simplicity that the drive/motor transfer function has unity gain, it is trivial to show that the closed-loop system state-space model is,

$$\begin{aligned}\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\zeta} \\ \dot{\hat{\mathbf{x}}} \end{bmatrix} &= \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{H}\mathbf{C}^0\mathbf{C} & \mathbf{B}\mathbf{E} & \mathbf{B}G_{ff}\mathbf{C}_f^0 \\ -\Gamma\mathbf{C}^0\mathbf{C} & \Phi & \mathbf{0} \\ \mathbf{L}\mathbf{C} - \mathbf{B}_f\mathbf{H}\mathbf{C}^0\mathbf{C} & \mathbf{B}_f\mathbf{E} & \mathbf{A}_f - \mathbf{L}\mathbf{C}_f + \mathbf{B}_fG_{ff}\mathbf{C}_f^0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \zeta \\ \hat{\mathbf{x}} \end{bmatrix} \\ &+ \begin{bmatrix} \mathbf{B}\mathbf{H} & -\mathbf{B}\mathbf{H}\mathbf{C}^0\mathbf{D} & -\mathbf{B} \\ \Gamma & -\Gamma\mathbf{C}^0\mathbf{D} & \mathbf{0} \\ \mathbf{B}_f\mathbf{H} & \mathbf{K}\mathbf{D} - \mathbf{B}_f\mathbf{H}\mathbf{C}^0\mathbf{D} & \mathbf{0} \end{bmatrix} \begin{bmatrix} r \\ \omega_b \\ T_f \end{bmatrix}\end{aligned}\quad (23)$$

The closed-loop system state vector then contains the plant, controller and filter states and the input variables are those of the reference signal, base rate and friction torque. Addition of the friction torque estimate to the control signal is apparent (closed-loop system matrix element (1,3)), but it also appears in the filter system matrix (3,3). Ordinarily in an LQG design, the filter dynamic matrix is $\mathbf{A}_f - \mathbf{L}\mathbf{C}_f$. Altering the input to the Kalman filter to $\mathbf{u}_f = \mathbf{u}_c$ recovers this form. Now, however, the input to the filter model is different to the actual plant input containing the filter torque estimate. This slight change to the architecture of the feedback/observer structure alters the dynamic response of the closed-loop system markedly, as can be seen in Figure 9. With exact disturbance matching the compliance transfer function has zero magnitude at DC, yielding theoretically infinite stiffness. This will be denoted the “*old*” control-signal configuration with filter input given by Eq. (22) defined with unity feedforward gain G_{ff} . This is illustrated in Figure 9 by the solid thick line. Similarly, the “*new*” control-signal architecture is shown by the thick dashed line in Figure 9. This new architecture always maintains a finite stiffness at low frequency meaning that it exhibits poorer disturbance rejection properties in this region. However, this is not indicative of actual disturbance rejection performance on a real transmitter, as it is unlikely that the rate loop would be implemented without an integrating position loop, from an image tracker for example, which will introduce additional stiffness at low frequencies. Also shown in Figure 9, it is impossible to improve on the classical estimator performance by

utilizing the feedforward gain term, as this degrades the compliance significantly (c.f. new architecture where compliance is improved). The effect of employing the new architecture is to sacrifice performance when the estimated friction torque exactly matches the actual friction torque (highly unlikely in practice) to obtain an additional design degree of freedom. Finally, and perhaps most importantly, sightline stiffness is increased around the bandwidth of the controller, which will result in an improved disturbance rejection in this region. This is significant because many of the exogenous disturbances that act upon an airborne system such as atmospheric loads, cable winding torques or other mechanical imperfections generally have spectra concentrated around 10 Hz. The modified control architecture should therefore exhibit better rejection of disturbances that are not captured adequately by the disturbance model in the filter.

Simulation Results

Estimator convergence. The first test of each friction observer was to check the convergence properties across the problematic jitter frequency range (2 – 60 Hz). Time histories of the simulated friction torque produced by the LuGre model and the estimated torque from each observer were compared. Figure 10 shows the time histories for a 2 Hz sinusoidal base motion input for the linear KF, Coulomb and Dahl model observers, respectively. Note that this result corresponds to the open-loop case ($G_{ff} = 0$) and these results are valid for either controller architecture.

It is immediately apparent that while the torque estimates from all three Kalman filters do converge to the true friction torque, there is a marked difference between both the tracking accuracy and noise levels in the linear filter estimate compared to the two EKFs. This is because the linear filter has no information about the switching characteristic of bearing friction torque around zero velocity. The linear filter is simply tracking the state estimates as fast as it can given the observer bandwidth as determined by the Kalman gain $\mathbf{L}$. Higher observer gain leads to closer tracking of the friction switch at the expense of greater noise passed through from the measurement signals. Both EKF formulations do not suffer from this defect as they each explicitly predict the friction torque reversal characteristics, albeit to a reduced fidelity.

Unlike the linear Kalman filter, significant care has to be taken when selecting the structure of the process noise matrix $\mathbf{G}(\mathbf{x})$ due to the torque reversal in both EKF models and the high bearing stiffness gain required by the Dahl model. Consider first the Dahl model EKF and assume that $\mathbf{G}(\mathbf{x})$ is constant for $t \in [0, \infty)$. The estimator will converge assuming that the base motion is unidirectional. However, if the base motion changes direction, divergence in the torque estimation channel of the filter may occur in the first instance, but quickly expands to the entire filter state vector. This possible destabilization is related to the innovations signals undergoing a sign change as might be expected, but also to $(\omega_i - \omega_b)$ changing sign inside $\mathbf{A}(\mathbf{x})$. An example is shown in Figure 11, where the divergence of $\hat{\mathbf{F}}_c$ can be clearly seen after 0.25 s. At 0.5 seconds the base motion sinusoid causes another velocity reversal that realigns the filter sign and the friction estimate converges again. This pattern is repeated until either the estimate converges to the true values of the state or the filter becomes globally unstable. In this case, stability is intrinsically coupled to the base

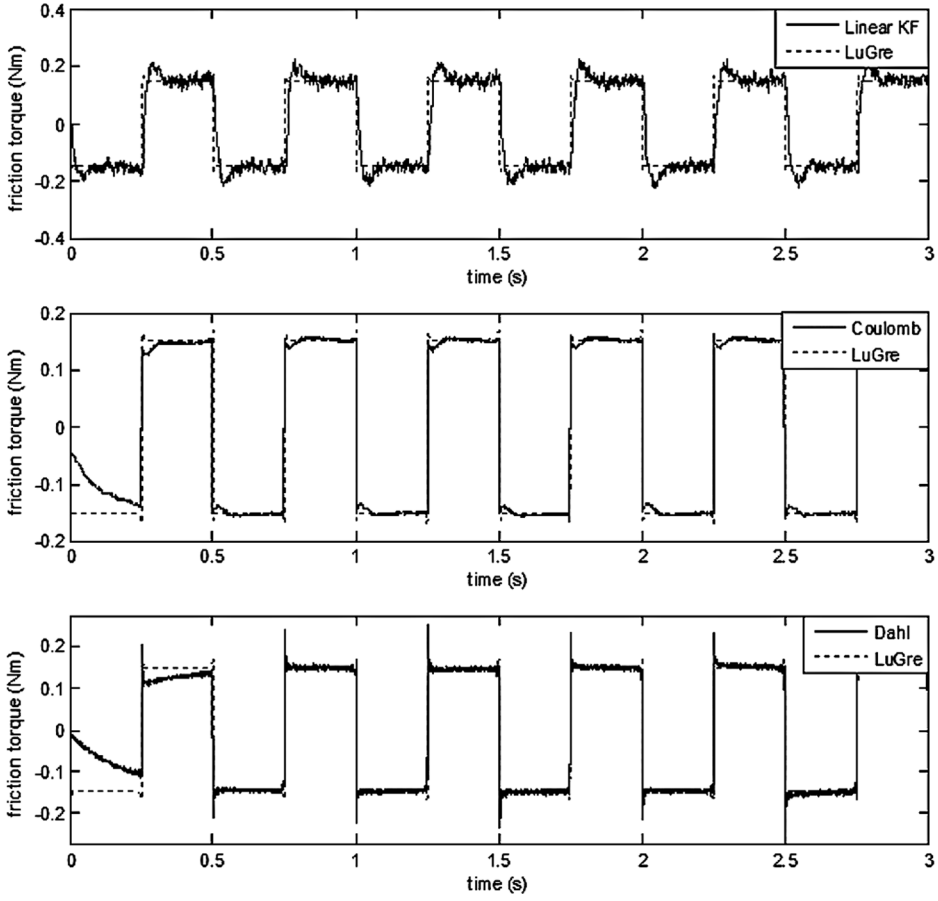


Figure 10. EKF torque time histories. The dashed lines are the friction torque profiles generated by the Lu-Gre model in the simulation. The solid lines show the time histories of the three filters. The noise sensitivity of the linear filter is clear from the top chart.

motion state, which is undesirable. Also shown in Figure 11 is the time series of the Coulomb friction state estimate for a Dahl filter with rectification on the measurement stream, achieved by setting,

$$\mathbf{G}(x) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ -2*\text{sgn}(\hat{\omega}_i - \hat{\omega}_b) & 0 \end{bmatrix} \quad (24)$$

The $\text{sgn}(\hat{\omega}_i - \hat{\omega}_b)$ function rectification in the error covariance iteration stream ensures that the Kalman gain $\mathbf{L}(x)$ acts in the correct direction, even with significant gain change in the model. However, slight this modification is to the EKF, it is of critical importance to this application.

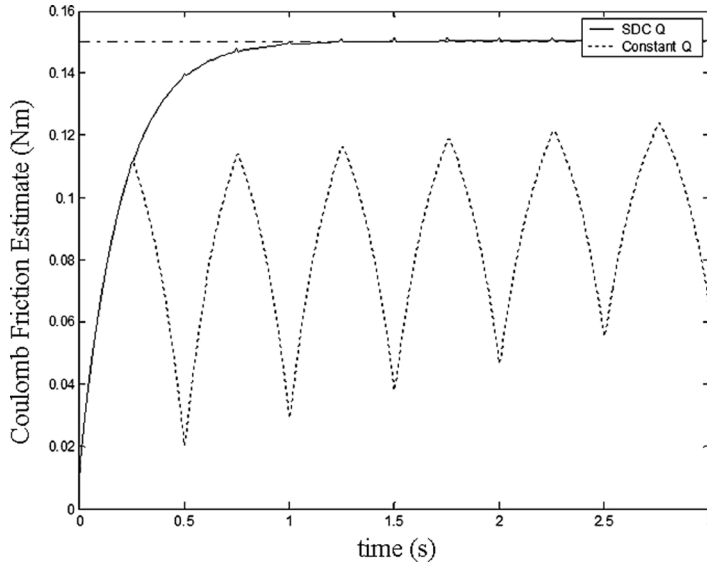


Figure 11. Coulomb friction convergence comparison with/without rectification.

Friction Compensation Performance

Frame-to-frame jitter. The desired performance of the friction compensator can be neatly encapsulated by construction of a compliance plot using the Lu-Gre model to provide the friction torque and measuring the RMS jitter levels in the LoS to sinusoidal base motion. The base motion sinusoid has a peak angular acceleration of 10 rad/s/s, a value selected to exceed expected in-service levels, with the following discrete frequency vector, $\omega_b = [2\ 5\ 10\ 20\ 30\ 40\ 50\ 60]$ Hz. Each simulation is run over 10 seconds such that sufficient data is generated to compute a stable, converged RMS. In addition, each run was performed with the noise model on the FOGs activated using parameters obtained from the manufacturers specification and data sheets.

It is clear from Figure 12 that all six friction compensators have successfully reduced the LoS jitter. The peak jitter value has been reduced from 92 μ rad to under 30 μ rad for all cases, a three-fold improvement over the baseline controller. The compliance frequency response shapes of Figure 9 are also evident with the new observer architecture settling to a peak of around 28 μ rad rms at 2 Hz base input frequency and the old observer architecture moving to the noise floor. It is also clear that using the nonlinear models improves the compensation performance. This is because the explicit switching included in the nonlinear filter models mean a lower bandwidth estimator can be used, leading to less noise in the signal estimate. Model sensitivity is most clearly demonstrated in the compensator comprising a linear Kalman Filter with the zero-DC gain (old) controller architecture, which exhibits significantly poorer performance between 5 and 30 Hz. This is most likely due to a bode integral effect (Skogestad and Postlethwaite 1996) because the old configuration imposes a constraint of zero compliance at DC, this energy must appear somewhere in the spectra. The linear Kalman filter model has a finite bandwidth, which meets the inertia

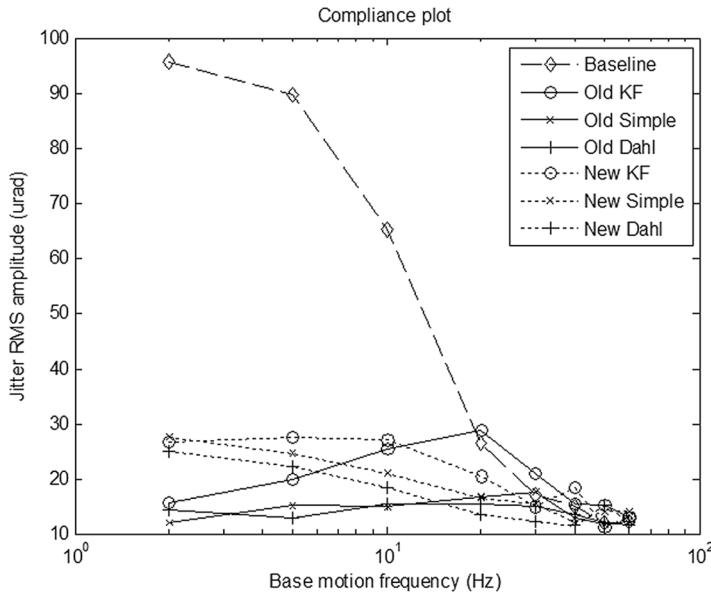


Figure 12. Comparison of compliance plots for friction compensation controllers with $G_{ff} = 3$.

line of the gimbal in the frequency range of interest. The switching function in the nonlinear models moves some of this energy to high frequency (the exact frequency a function of the low-speed dynamics), keeping the compliance low. As the new controller does not impose a zero DC compliance, this effect does not manifest. A final point to note is the sensitivity of each approach to modelling inaccuracies, in particular inaccurate estimation of the gimbal inertia. The results of Figure were obtained with the inertia exactly matched in the old configuration and a factor of four mismatch in the new. The performance of the old architecture would deteriorate quickly as the simulation and model inertias diverge, but this is not the case with the new. This result illustrates the potential for the new configuration to be more robust than the old to modelling inaccuracies.

Image integration jitter. The compliance results presented in Figure clearly suggest that the old disturbance controller architecture is the preferred design. As discussed however, image quality is strongly influenced by sightline jitter during the imager integration dwell-time and it is the contention of this article that image quality can be improved by shaping in addition to minimising sightline jitter. Differences in sightline jitter profiles are shown in Figure 12 for both compensator configurations at two base frequencies (2 and 60 Hz). These two frequencies were chosen to illustrate the differences in jitter shape between the two controller architectures at 1) low frequency where friction torque dominates and 2) high frequency where sensor noise dominates. The intermediate frequencies blend smoothly between these endpoints.

A visual comparison of plots (a) and (c) in Figure 13 clearly suggests a difference in structure (frequency content). Peak deviations are comparable in amplitude, fluctuating between $\pm 50 \mu\text{rad}$ but where in (a) the large-scale deviations occur as a series of spikes superimposed on an underlying sinusoid, in (c) the jitter time series

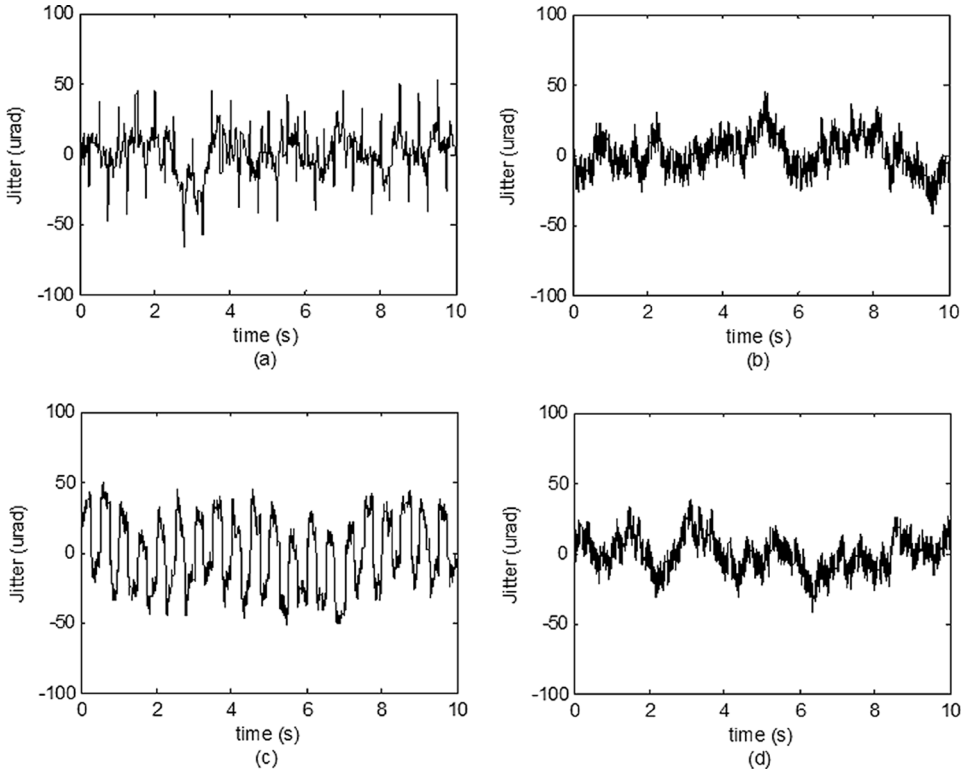


Figure 13. Jitter time histories *a)* old compensator, 2 Hz base motion, *b)* old compensator, 60 Hz base motion, *c)* new compensator, 2 Hz base motion, and *d)* new compensator, 60 Hz base motion.

has much more of a square-wave profile. This is reflected in the standard deviations of both signals which are $18\mu\text{rads}$ for *(a)* and $27\mu\text{rads}$ for *(c)*. The jitter levels at 60 Hz are identical with equivalent standard deviations and almost perfect normal probability distribution function (*pdf*) as noise from the FOGs dominates the jitter signal at this frequency.

To include the effect of finite imager integration time on the sightline jitter statistics, the jitter signals *(a)* and *(c)* were passed through a digital sampling filter of the form,

$$y_n(t) = \begin{cases} x(\tau) & t_n \leq \tau \leq t_n + \Delta\tau \\ 0 & \text{otherwise} \end{cases} \quad n = 0, 1, 2, \dots \quad (25)$$

where $\Delta\tau$ is the imager dwell time and t_n is the time at which the n^{th} imager integration occurs. The standard deviation of each $y_n(t)$ is computed and the resulting vector of sightline motions passed to the MATLAB statistics toolbox to compute the closest pdf. The results of this exercise are shown in Figure 14. The “best-fit” normal distributions for both the old and new controller configurations are shown. The theoretically perfect distribution would be a dirac delta function around the

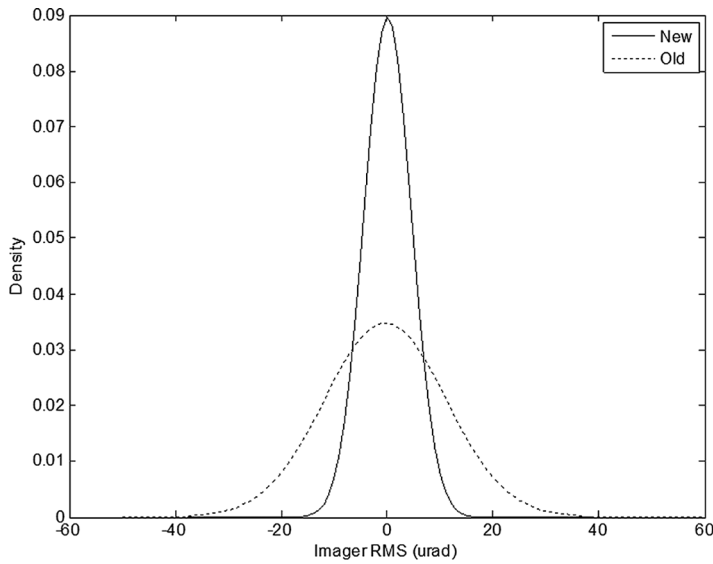


Figure 14. Imager integration probability distributions.

mean (zero) value. From Figure 14 the new controller configuration performs much better at minimizing the dwell-time jitter, even though the unfiltered rms value is 50% higher. This result is due entirely to the shaping of the sightline jitter time history.

CONCLUSIONS

Several conclusions can be drawn from the work presented here, but the two main contributions will be summarized first. It has been shown that an EKF can be designed to estimate the dynamic response of bearing friction using a model that includes the hard nonlinearities around the velocity reversal. The filter provided a much-improved estimate of the friction torque, especially across torque reversal than was achievable with a linear state estimator. Central to the success was the inclusion of a state-dependant nonlinearity in the velocity reversal signum, placed in the error covariance iteration stream, thus ensuring synchronization between both model prediction and filter gain calculation steps. The experiments were conducted at single base frequencies in accordance with standard transfer function analysis procedure, but as the filters do not require base frequency information in the estimation process, the technique should be valid for realistic, multiple-frequency disturbances.

By changing the architecture of the controller, performance was sacrificed for the exact friction estimation case to obtain significantly improved robustness to errors in the filter model. However, recovery of friction compensation performance was shown to be possible through the inclusion of a gain term on the filter torque estimate signal. Although the validation exercise and results calculated by the simulation were obtained by applying the maximum base motion acceleration, the robustness exhibited by the new controller configuration should be sufficient to handle any setpoint nonlinearity.

Excellent improvement in stabilization was shown to be achievable with fast estimation and torque feedforward gain for each of the models used. This result suggested that Coulomb friction dominates jitter in an EO system track loop; therefore, there is little expected benefit to be obtained from including higher-order friction model terms in the EKF such as the Stribeck velocities for example. However, the system under test has low-noise sensors, which allowed the linear KF bandwidth to be set relatively high. For systems with noisier measurements, the EKF would be noticeably superior to the linear KF because the switching inside the model and Kalman gain update equations leads to better tracking with much less noise sensitivity.

Finally, and most importantly, a new approach to the design of sightline stabilization controllers was presented. This method was based upon the classical approach to friction compensation using an output-injection observer to estimate friction torque, but by altering the signal pathways in the controller both the shape and standard deviation of the jitter were manipulated. The sampling and integration characteristics of the imager coupled with the controller to yield a stabilization strategy that improves upon the classical approach. Although this new controller was a simple modification of a classical approach, the excellent results suggest that a signal-shaping controller is the optimal solution to the sightline stabilization problem.

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